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VIOLENCE DETECTION IN VIDEOS USING HUMAN ACTION RECOGNITION

Graduation Project Interim Report

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Statement of Authenticity

I/we hereby declare that in this study

1. all the content influenced from external references are cited clearly and in detail,
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İstanbul, May 2026

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Violence Detection in Videos Using Human Action Recognition

(SUMMARY)

Automated violence detection in surveillance video is a critical task for public-safety systems. The proliferation of closed-circuit cameras in public spaces creates an urgent need for reliable, real-time detection systems that can alert security personnel without requiring constant human monitoring. While manual review remains the norm, the sheer volume of surveillance footage makes it impractical, strongly motivating fully automated solutions.

Existing approaches to automated violence detection rely predominantly on appearance-based 3D convolutional networks or optical-flow features. These methods, while effective, leave body-pose dynamics largely unexplored as a discriminative cue. Optical-flow computation introduces significant overhead in both computation and storage, and appearance-only models can be confused by non-violent scenes with high visual energy such as sports events or crowded celebrations.

This project presents a three-stream late-fusion framework that addresses these gaps by combining complementary representations of the same video. The first stream is an RGB stream based on a fine-tuned R3D-18 backbone pre-trained on Kinetics-400. The second and third streams are skeleton-based: a joint stream that models absolute keypoint coordinates and a bone stream that models relative limb vectors, both implemented using Two-Stream Adaptive Graph Convolutional Networks (2s-AGCN). Pose keypoints are extracted per-frame using YOLO11n-pose, retaining the two most salient persons per frame based on a compound score of bounding-box area and mean keypoint confidence.

The RGB stream employs a two-phase training strategy with gradual backbone unfreezing combined with whole-video random frame sampling. This strategy raises single-stream accuracy from 83.0% to 88.0% on the RWF-2000 benchmark. The two skeleton streams are initialised from weights pre-trained on the NTU RGB+D dataset and fine-tuned on RWF-2000 pose annotations using the MMAction2 framework. Stream-level softmax scores are combined via a weighted sum whose coefficients are determined by exhaustive grid search on the validation partition. The optimal weights found are 0.45 for the RGB stream, 0.25 for the joint stream, and 0.30 for the bone stream.

On the RWF-2000 benchmark (2,000 clips, 1,600 training / 400 validation), the proposed three-stream fusion achieves 89.25% accuracy and an F1-score of 0.892, surpassing the dataset's original Flow Gated Network baseline of 87.25% by 2 percentage points while using no optical flow. An ablation study confirms that every stream combination improves over its constituent streams and that the bone stream is the more informative skeleton modality, outperforming the joint stream both individually and in pairwise fusion with RGB.

Zero-shot transfer to the Real-Life Violence Situations (RLVS) dataset, containing 1,951 clips not seen during training, yields 81.19% accuracy, F1 = 0.838, and 95% violence recall. The high recall demonstrates strong cross-domain generalisation and underlines the model's

suitability for safety-critical deployments where missed detections are costly. The framework's modularity, computational efficiency relative to optical-flow pipelines, and inherently privacy-preserving skeleton representations make it a promising foundation for practical surveillance applications.

İnsan Hareketi Tanıma Tabanlı Videolarda Şiddet Tespiti

(ÖZET)

Gözetim videolarında otomatik şiddet tespiti, kamusal güvenlik sistemleri için kritik öneme sahip bir görevdir. Kapalı devre kameraların yaygınlaşması, sürekli insan gözetimi gerektirmeksizin güvenlik personelini uyarabilecek güvenilir, gerçek zamanlı tespit sistemlerine yönelik acil bir ihtiyaç doğurmaktadır. Manuel inceleme yaygın uygulama olmaya devam etse de, gözetim görüntülerinin çok büyük hacmi bunu pratik olmaktan çıkarmakta ve tam otomasyona olan ihtiyacı güçlü biçimde gerekçelendirmektedir.

Mevcut otomatik şiddet tespit yaklaşımları büyük ölçüde görünüme dayalı 3B evrişimli ağlara veya optik akış özelliklerine dayanmaktadır. Bu yöntemler etkin olmakla birlikte, vücut pozu dinamiklerini ayırt edici bir ipucu olarak büyük ölçüde göz ardı etmektedir. Optik akış hesaplama hem hesaplama hem de depolama açısından önemli ek yük getirmekte; yalnızca görünüme dayalı modeller ise spor etkinlikleri ya da kalabalık kutlamalar gibi görsel enerjisi yüksek şiddet içermeyen sahnelerde yanılabilmektedir.

Bu proje, söz konusu eksiklikleri gidermek amacıyla üç akışlı bir geç füzyon çerçevesi sunmaktadır. Birinci akış, Kinetics-400 üzerinde ön eğitimli ve ince ayarlı R3D-18 omurgasına dayanan bir RGB akışıdır. İkinci ve üçüncü akışlar ise iskelet tabanlıdır: biri mutlak eklem koordinatlarını modelleyen bir eklem akışı, diğeri göreli uzuv vektörlerini modelleyen bir kemik akışıdır; her ikisi de İki Akışlı Uyarlanabilir Grafik Evrişimli Ağ (2s-AGCN) mimarisi kullanılarak gerçekleştirilmiştir. Poz anahtar noktaları her kare için YOLO11n-pose kullanılarak çıkarılmakta; sınırlayıcı kutu alanı ile ortalama anahtar nokta güvenilirliğinin çarpımına dayalı bileşik bir puanla kareye ait en önemli iki kişi seçilmektedir.

RGB akışı, aşamalı omurga çözme ve tüm videodan rastgele kare örneklemesini birleştiren iki aşamalı bir eğitim stratejisi kullanmaktadır. Bu strateji, RWF-2000 kıyaslama setinde tek akış doğruluğunu %83,0'dan %88,0'a yükseltmektedir. Her iki iskelet akışı da NTU RGB+D veri seti üzerinde ön eğitilmiş ağırlıklarla başlatılmakta ve MMAction2 çerçevesiyle RWF-2000 poz açıklamaları üzerinde ince ayar yapılmaktadır. Akış düzeyindeki softmax skorları, doğrulama bölümünde kapsamlı ızgara arama yoluyla belirlenen katsayılara sahip ağırlıklı bir toplam aracılığıyla birleştirilmektedir.

RWF-2000 kıyaslama setinde önerilen üç akışlı füzyon %89,25 doğruluk ve F1 = 0,892 elde etmekte; optik akış kullanmaksızın veri setinin orijinal Akış Geçitli Ağ tabanını 2 yüzde puanı geride bırakmaktadır. Sıfır atışlı aktarım değerlendirmesinde ise RLVS veri setinde %81,19 doğruluk, F1 = 0,838 ve %95 şiddet geri çağırım oranı elde edilmiştir. Bu yüksek geri çağırım oranı, modelin alanlar arası genelleşme kapasitesini ortaya koymakta ve güvenlik açısından kritik uygulamalar için uygunluğunu vurgulamaktadır.

Contents

1 Introduction and Project Summary

The proliferation of closed-circuit surveillance cameras in public spaces has created an urgent need for reliable, real-time violence detection systems. Security personnel cannot monitor every camera continuously, making automated detection indispensable for rapid incident response and public safety management.

Early computational approaches encoded motion through hand-crafted descriptors such as Histogram of Oriented Optical Flow (HOOF) [12] or spatio-temporal interest points. The advent of deep learning shifted the field toward 3D convolutional networks [4] that jointly model appearance and short-range motion, and two-stream networks [5] that separately process RGB frames and optical-flow fields before fusing their predictions. The RWF-2000 dataset [1], a large-scale benchmark of real surveillance clips, established a competitive baseline of 87.25% accuracy using a Flow Gated Network that combines optical-flow attention with spatial appearance features.

Despite these advances, two important cues remain underexplored in the violence-detection literature. First, body pose provides a semantically interpretable representation of human motion that is inherently invariant to background clutter and illumination changes. Second, bone vectors—directed differences between adjacent keypoints—capture limb-level kinematics that complement raw joint positions. Graph Convolutional Networks (GCNs) designed for skeleton-based action recognition, most notably the Two-Stream Adaptive GCN (2s-AGCN) [2], exploit both representations effectively on general action benchmarks but have seldom been applied to the specific challenge of violence detection in unconstrained surveillance footage.

This project presents a modular three-stream late-fusion pipeline that addresses these gaps. The framework combines: (i) an RGB stream based on a fine-tuned R3D-18 backbone [3]; (ii) a joint stream modelling absolute keypoint coordinates with 2s-AGCN; and (iii) a bone stream modelling relative limb vectors with the same architecture. Pose keypoints are extracted per-frame using YOLO11n-pose [7], retaining the two most salient persons. Stream-level softmax scores are combined by a weighted sum optimised via exhaustive grid search.

On the RWF-2000 benchmark our best configuration achieves 89.25% accuracy and F1 = 0.892, surpassing the dataset's original baseline by 2 percentage points without using optical flow. Zero-shot transfer to the Real-Life Violence Situations (RLVS) dataset yields 81.19% accuracy with 95% violence recall, demonstrating strong cross-domain generalisation.

1.1 Engineering Standards and Multiple Constraints

The design of this system was shaped by a set of real-world engineering constraints that influenced every architectural decision, from model selection to deployment strategy. Engineering design is an iterative, creative process that converts available resources into solutions while satisfying competing requirements. The constraints encountered in this project span computational, functional, privacy, and operational dimensions.

All training and experimentation were conducted on Google Colab with access to an NVIDIA A100/T4 GPU. This cloud-based constraint ruled out architectures with very high memory footprints (e.g., large transformer-based video models) and motivated the choice of R3D-18 over heavier alternatives such as I3D or Video Swin Transformer. The 3-stream modular design is itself a response to this constraint: training each stream independently requires far less peak GPU memory than an end-to-end joint model would. Pose extraction was performed offline in a preprocessing pass rather than on-the-fly during training, reducing the per-epoch computational burden at the cost of disk storage for the annotation pickle files.

A violence detection system deployed in a real public-safety context must satisfy asymmetric reliability requirements: missed violent detections (false negatives) carry a higher operational cost than false alarms, since a false alarm triggers a human review while a missed detection results in no intervention. This constraint guided both the choice of evaluation metric (violence recall as a primary indicator alongside accuracy and F1) and the calibration of the fusion weights. The grid-search-optimised weights (w_RGB = 0.45, w_joint = 0.25, w_bone = 0.30) were selected to maximise F1 on the validation set, with the additional observation that the system achieves 95% violence recall on the unseen RLVS dataset. The minimum accuracy target of 80% set at project inception was exceeded on both datasets (89.25% on RWF-2000, 81.19% on RLVS).

Surveillance footage presents severe environmental variability: low resolution, motion blur, occlusion, variable illumination, and wide-angle distortion. Appearance-only models are vulnerable to these conditions because their learned texture features are tied to scene context. The inclusion of skeleton streams directly addresses this robustness constraint: pose-based representations are invariant to background content, illumination changes, and camera viewpoint, making them naturally suited to the unpredictable conditions of real CCTV installations. The bone stream specifically addresses scale and position invariance, since directed limb vectors are unaffected by global body translation across the frame. The zero-shot RLVS evaluation (81.19% accuracy without any retraining) provides empirical evidence that the pose-centric design promotes cross-domain robustness.

Automated surveillance systems raise significant privacy concerns, as continuous video analysis may facilitate mass identification or tracking of individuals. The skeleton-based streams of the proposed system provide a natural path toward privacy-preserving deployment: once keypoint coordinates are extracted per-frame by YOLO11n-pose, the raw pixel frames are no longer required by the GCN inference pipeline. A production deployment could therefore perform on-device pose extraction and immediately discard the raw video, transmitting only compact skeleton tensors to a server for classification. This architectural affordance substantially reduces the risk of identity disclosure while preserving the motion signals needed for violence recognition, aligning the system with privacy-by-design principles increasingly required by data protection regulations.

The modular late-fusion architecture was chosen in part to satisfy extensibility and maintainability constraints. Because each stream is trained independently and integrated only at the score level, any individual stream can be replaced or upgraded without retraining the others. For example, the R3D-18 RGB stream could be swapped for a more powerful backbone, or the YOLO11n-pose extractor replaced with a higher-accuracy model, with the only re-tuning step being a new grid search over fusion weights on the validation set. This design significantly reduces the maintenance burden compared to an end-to-end jointly trained model, where any architectural change would require full retraining. The system was implemented entirely in Python using standard open-source libraries (PyTorch, MMAction2, Ultralytics YOLO), ensuring long-term maintainability and reproducibility without proprietary dependencies.

2 Comparative Literature Survey

2.1 Appearance-Based Violence Detection

3D CNNs such as C3D [4] and ResNet-based variants learn spatiotemporal features in a single forward pass over short video clips. Cheng et al. [1] proposed the Flow Gated Network, which uses optical flow as an attention mask over appearance features and established the 87.25% baseline on RWF-2000. Early optical-flow-based methods such as Hassner et al. [15], who introduced the Violent Flows descriptor for crowd violence detection, and Deniz et al. [16], who proposed a fast optical-flow pipeline for real-time analysis, demonstrated that motion dynamics carry strong discriminative signal. However, these hand-crafted descriptors lack the representational capacity of learned features.

More recent deep learning approaches have explored feature-level fusion strategies. Bin Jahlan and Elrefaei [17] proposed a CNN-based deep feature fusion method that combines feature maps from multiple network layers, achieving accuracy improvements on standard benchmarks. Sudhakaran and Lanz [19] demonstrated that Convolutional LSTM architectures effectively model temporal dependencies in violent scenes, and Asad et al. [20] extended this with a spatio-temporal ConvLSTM formulation. Transformer-based methods [13], [14] obtain higher accuracy on general action recognition benchmarks but incur substantially higher computational cost, limiting deployment on edge surveillance hardware.

Our work differs from these appearance-centric methods by incorporating body-pose dynamics as complementary streams, allowing the system to exploit structural motion cues not captured by pixel-level features alone.

2.2 Skeleton-Based Action Recognition

Graph Convolutional Networks model the human body as a graph whose nodes are joints and edges are bones. ST-GCN [6] pioneered this approach with a fixed topology; 2s-AGCN [2] extended it by learning adaptive adjacency matrices and processing joint and bone streams independently. Li et al. [21] further proposed Actional-Structural GCN, which separately models structural and behavioural joint relationships, while Si et al. [22] integrated GCN and LSTM with an attention mechanism to improve temporal dependency modelling.

The most closely related skeleton-based violence detection work is Zhao et al. [23], who proposed a Dual-Stream GCN architecture operating exclusively on skeleton representations. While their method demonstrates the value of graph-based pose modelling, it does not incorporate any appearance-based RGB stream, leaving complementary visual context unexploited. Our work addresses this gap by fusing skeleton dynamics with an RGB backbone, allowing the model to leverage both structural motion cues and appearance priors.

2.3 Multi-Stream Fusion Strategies

Late fusion of independently trained streams is a well-established strategy in video understanding [5], [11]. Score-level fusion with learned or searched weights consistently outperforms single streams while preserving modularity. Feichtenhofer et al. [11] demonstrated that fusion architecture choices significantly affect recognition accuracy. Our work adopts this paradigm, fusing RGB appearance with skeleton dynamics via a weighted softmax combination whose coefficients are optimised by exhaustive grid search on the validation partition.

3 Developed Approach and System Model

The proposed framework is a three-stream late-fusion pipeline. Each stream is trained independently, and the final fusion layer requires only a single validation-set grid search to calibrate stream weights. This modularity makes it straightforward to replace or upgrade any individual stream without retraining the entire system.

3.1 Data Model

Two datasets are used in this work:

RWF-2000 [1]: Contains 2,000 five-second surveillance clips balanced between Fight and NonFight categories, collected from YouTube. The official split uses 1,600 clips for training and 400 for validation. All clips are resized to 320×240 pixels at 30 fps. This dataset serves as the primary benchmark due to its scale, real-world conditions, and the availability of a well-established baseline.

RLVS [10]: The Real-Life Violence Situations dataset comprises 1,951 clips sourced from the internet, balanced between violent and non-violent content. RLVS is used exclusively for zero-shot cross-dataset evaluation; the model is trained only on RWF-2000 and tested on all 1,951 RLVS clips without any adaptation.

Pose annotations are generated by processing each video frame-by-frame with YOLO11n-pose at its native resolution with a detection confidence threshold of 0.25. For each detected person, a salience score is computed as the product of bounding-box area and mean keypoint confidence, and the top-2 persons per frame are retained. Frames with no detections are filled with zero tensors. The resulting per-video tensor has shape (T, 2, 17, 3), where T is the frame count, 2 is the maximum number of persons, 17 is the number of COCO keypoints, and 3 comprises (x, y, confidence). For MMAction2 [8] compatibility, keypoint and confidence arrays are transposed to shape (M, T, 17, 2) and (M, T, 17) respectively and serialised to a pickle annotation file.

3.2 Structural Model

RGB Stream (R3D-18)

The RGB stream is built on R3D-18 [3], a 3D residual network pre-trained on Kinetics-400. The final fully connected layer is replaced with a two-class head. Training follows a three-phase gradual-unfreezing strategy:

- Phase 1 (epochs 1–5): the backbone is frozen and only the classification head is trained.
- Phase 2 (epochs 6–12): the last residual block is unfrozen.
- Phase 3 (epochs 13–20): the entire network is fine-tuned with a reduced learning rate (10^{-5}).

The above describes the final training strategy (v2). An earlier version (v1) used a simpler two-phase approach: the backbone was first fully frozen, then all layers were unfrozen at once, and frames were sampled as a consecutive 16-frame window from a random starting point within the video. The final checkpoint (last epoch) was saved regardless of validation performance. This produced a single-stream accuracy of 83.00%.

The v2 strategy differs in three key ways. First, frame sampling was changed from consecutive windowing to random selection across the entire video duration. Surveillance violence clips often contain sparse but highly informative moments distributed throughout the clip; global random sampling substantially increases the probability of capturing these discriminative interactions during each training iteration, effectively acting as temporal data augmentation. Second, the unfreezing schedule was extended to three gradual phases, allowing the classification head to stabilise before progressively deeper spatiotemporal features are updated. This prevents large gradient updates from disrupting the pretrained Kinetics-400 representations early in training. Third, the best validation checkpoint is saved rather than the last epoch, guarding against overfitting in the final training phase when the learning rate is lowest and the model is most susceptible to memorising training samples.

These changes together raised single-stream RGB accuracy from 83.00% to 88.00%, and also improved the RGB+Joint fusion from 85.00% to 88.00%. Table 3.1 summarises the differences between the two training configurations and their impact on accuracy.

Table 3.1: Comparison of RGB training strategies v1 and v2.

	v1	**v2**
Frame Sampling	Consecutive window	Random (whole video)
Unfreezing	Freeze then unfreeze all	3 gradual phases
Checkpoint	Last epoch	Best validation epoch
RGB only accuracy	83.00%	**88.00%**
RGB + Joint accuracy	85.00%	**88.00%**

Skeleton Streams (2s-AGCN).

Two parallel 2s-AGCN models [2] process the skeleton annotations independently. The joint stream uses 2-D pixel coordinates (x, y) of each COCO keypoint as node features; the graph topology follows the standard 17-joint skeleton and the adaptive adjacency mechanism learns inter-joint relationships beyond the physical skeleton. The bone stream computes each bone vector as the directed difference between a joint and its parent in the kinematic tree (e.g., b_LeftShoulder = j_LeftElbow − j_LeftShoulder), capturing limb orientation and speed independently of absolute body position. Both models are initialised from weights pre-trained on NTU RGB+D (cross-subject split) and fine-tuned for 80 epochs on the RWF-2000 pose annotations using MMAction2's standard pipeline.

Late Fusion

Given softmax probability vectors p_RGB, p_joint, and p_bone for a test video, the fused prediction is computed as:

$$p\_fused = w\_RGB \cdot p\_RGB + w\_joint \cdot p\_joint + w\_bone \cdot p\_bone$$

subject to w_RGB + w_joint + w_bone = 1. The final class is arg max p_fused. Weight optimisation is performed via exhaustive grid search over all non-negative weight triples that sum to 1, stepping each weight in increments of 0.05 (231 candidate triples). The search is conducted on the RWF-2000 validation set. The optimal weights found are w_RGB = 0.45, w_joint = 0.25, and w_bone = 0.30.

3.3 Dynamic Model

Figure 3.1 illustrates the overall processing pipeline. Input videos are processed in parallel by three independent streams. The RGB stream clips 16 frames uniformly at random from the full video and forwards them through R3D-18. Simultaneously, the pose extraction module applies YOLO11n-pose to every frame, selects the top-2 persons, and passes the resulting annotation tensors to two 2s-AGCN networks operating on joint and bone representations respectively. All three streams independently produce a two-class softmax probability vector (Fight, NonFight). These vectors are combined by the learned weighted sum, and the class with the highest fused probability is taken as the final prediction.

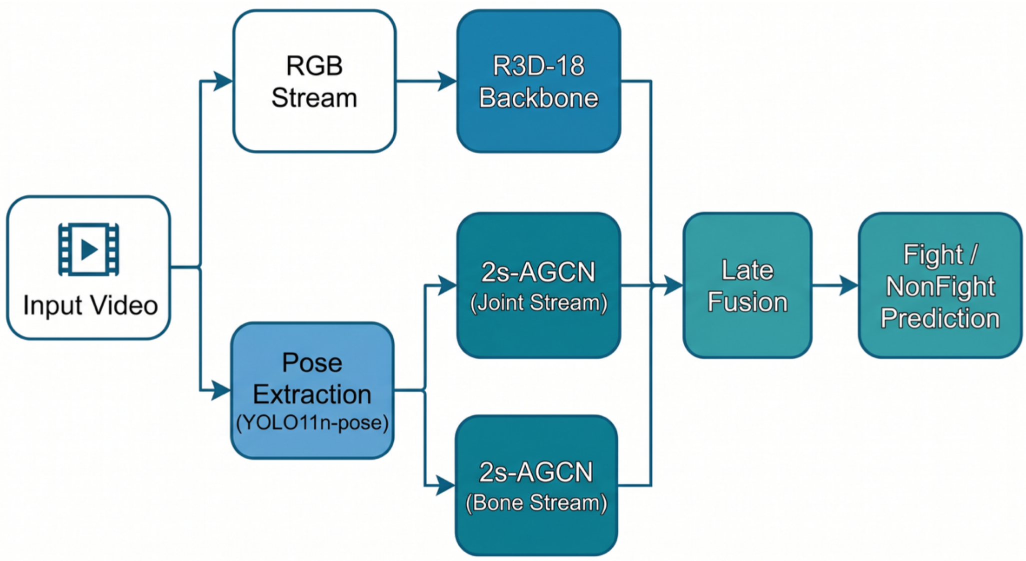

Figure 3.1: Overview of the proposed three-stream late-fusion pipeline. The RGB stream and skeleton streams process the input video in parallel, and their softmax outputs are combined through late fusion for the final Fight/NonFight prediction.

The pose extraction sub-module operates as follows: for each frame, YOLO11n-pose detects all persons above the confidence threshold of 0.25. Each detection is scored by the product of its bounding-box area and mean keypoint confidence. The top-2 scored detections are retained; missing slots are zero-padded. This salience criterion consistently selects the two subjects most directly involved in the scene's primary interaction, reducing the influence of partially visible bystanders.

To demonstrate the complete system, the three-stream pipeline was wrapped in a Gradio web interface. The interface accepts any video file as input, automatically runs the full pipeline (YOLO pose extraction, joint and bone 2s-AGCN streams, RGB R3D-18 stream, and late fusion), and outputs the final Fight/NonFight classification together with a confidence score.

This end-to-end prototype confirms that the modular design integrates cleanly into a single inference workflow.

4 Experimentation Environment and Experiment Design

All experiments are conducted on Google Colab using an NVIDIA A100 GPU. The following implementation details apply:

RGB Model: Trained with Adam optimiser (initial lr = 10^{-4}, decayed to 10^{-5} in Phase 3), batch size 8, for 20 epochs across three phases.

Skeleton Models: Trained with SGD (momentum 0.9, weight decay 5×10^{-4}) and a cosine annealing schedule over 80 epochs, batch size 16.

Pose Extraction: YOLO11n-pose processes each video frame-by-frame at native resolution with a confidence threshold of 0.25.

Fusion Grid Search: Exhaustive search over 231 candidate weight triples with step size 0.05, evaluated on the RWF-2000 validation set (400 clips).

The ablation study is designed to isolate the contribution of each component. Individual streams are evaluated first (RGB-only v1, RGB-only v2, joint-only, bone-only), followed by pairwise combinations (Joint+Bone, RGB+Joint, RGB+Bone) and finally the three-stream fusion with both manual and grid-searched weights. This design permits attribution of accuracy gains to specific design choices: the improved RGB training strategy, the inclusion of each skeleton stream, and the weight optimisation procedure.

Cross-dataset generalisation is assessed by applying the RWF-2000-trained model directly to all 1,951 RLVS clips without any fine-tuning or adaptation. This zero-shot protocol tests whether the model's reliance on body-pose dynamics promotes robustness to domain shift.

Sensitivity of the grid search is evaluated by fixing the RGB weight at values from 0.20 to 0.70 and splitting the remainder equally between joint and bone streams, producing the sensitivity curve reported in Table 5.3.

5 Comparative Evaluation and Discussion

Table 5.1 reports ablation results on the RWF-2000 validation set (400 clips). All figures are reported on the same fixed validation split to ensure comparability.

Table 5.1: Ablation results on RWF-2000 (validation set, 400 clips). †Grid-search weights; ‡manual weights.

Model / Stream	Accuracy (%)	F1
RGB only (v1)	83.00	0.843
RGB only (v2)	88.00	—
Joint only	80.75	—
Bone only	83.50	—
Joint + Bone	84.00	0.835
RGB + Joint	88.00	0.880
RGB + Bone	88.50	0.886
RGB + Joint + Bone ‡	87.25	0.878
RGB + Joint + Bone † (Ours)	**89.25**	**0.892**

Several observations follow from Table 5.1. First, the improved RGB training strategy (v2) alone recovers 5 percentage points over the baseline, confirming that random temporal sampling and gradual unfreezing are effective regularisers. Second, the bone stream (83.50%) consistently outperforms the joint stream (80.75%), suggesting that relative limb geometry is more discriminative than absolute joint positions. Third, every fusion combination improves over its constituent streams, with the three-stream configuration at grid-search weights reaching 89.25%. Fourth, the joint stream still adds value in the full three-stream fusion: comparing RGB+Bone (88.50%) with RGB+Joint+Bone at grid-search weights (89.25%) shows a 0.75-point gain, indicating that absolute joint positions carry complementary signal not fully redundant with bone vectors.

Table 5.2: Comparison with state-of-the-art methods on RWF-2000.

Method	Accuracy (%)	Input
Flow Gated Network [1]	87.25	RGB + Flow
I3D [9]	86.50	RGB + Flow
C3D [4]	82.75	RGB
Ours (3-stream, grid search)	**89.25**	**RGB + Pose**

Our method surpasses the Flow Gated Network baseline by 2 percentage points without using optical flow, which requires dense per-frame computation and substantial storage overhead. Using pose skeletons as a proxy for motion is both more computationally efficient and more interpretable.

Table 5.3: Zero-shot cross-dataset evaluation on RLVS (1,951 clips).

Metric	Value
Videos evaluated	1,951
Accuracy	81.19%
F1 Score	0.838
Violence Recall	95.0%
NonViolence Recall	66.7%

The model achieves 95% recall on violence, correctly flagging 950 out of 1,000 violent clips—a critical property for a safety system where missed detections are costly. The lower non-violence recall (67%) results in 317 false alarms, reflecting a natural bias in the fusion weights that prioritise sensitivity. This asymmetry is not undesirable in operational deployments: a false alarm triggers a human review, whereas a missed detection may result in no intervention at all.

The 317 false positives on RLVS are largely attributable to domain shift: RLVS NonViolence clips include gym workouts, martial arts demonstrations, and physically intense sports that are visually and kinematically similar to fight footage but absent from RWF-2000 training data. Threshold calibration or lightweight domain adaptation would likely substantially reduce these false alarms without degrading violence recall.

The sensitivity analysis of the grid search reveals that the accuracy curve is relatively flat in the range $w\_RGB \in [0.40, 0.55]$, suggesting the fusion is robust to exact RGB weight choice in this region. Performance degrades more steeply when the RGB stream is under-weighted ($w\_RGB < 0.35$), confirming its dominant role. A finer step size of 0.01 in the optimal range produced no further improvement, validating the resolution chosen.

6 Conclusion and Future Work

This project presented a three-stream late-fusion system for violence detection in surveillance video that combines an R3D-18 RGB backbone with dual 2s-AGCN skeleton streams operating on joint and bone representations. The key contributions and findings are:

- A two-phase RGB training strategy (gradual unfreezing + random temporal sampling) improved single-stream accuracy from 83% to 88%.
- The bone stream (83.50%) consistently outperformed the joint stream (80.75%), demonstrating that relative limb-level kinematics are more discriminative than absolute joint positions for violence recognition.
- Exhaustive grid search over fusion weights improved accuracy from 87.25% (manual weights) to 89.25% (optimal weights), surpassing the optical-flow baseline without requiring flow computation.
- Zero-shot transfer to RLVS achieved 81.19% accuracy with 95% violence recall, demonstrating strong cross-domain generalisation.

Several promising directions remain for future work. End-to-end joint training of all three streams with cross-modal contrastive supervision could further improve accuracy beyond independently trained streams. Temporal attention modules that focus on high-motion intervals would address the brittle uniform-sampling limitation, which currently causes brief violent actions occupying only a few frames to be missed. Domain adaptation strategies such as batch normalisation update or few-shot fine-tuning could reduce false alarms in new camera networks.

From a system perspective, real-time deployment would require batched YOLO inference and parallel stream execution on embedded hardware. Privacy-preserving deployments could benefit from on-device pose extraction followed by server-side GCN inference, discarding raw frames after pose extraction and substantially reducing identity-disclosure risk. Finally, surveillance-adapted pose estimation models trained on low-resolution, occluded footage would improve skeleton quality in the challenging conditions typical of wide-area CCTV installations.

7 References

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Appendix A: Modular System Implementation and Prototype Source Code

A.1. Spatio-Temporal Feature and Keypoint Extraction

The following script manages video intake, invokes a pre-trained **YOLO11n-pose** network, filters detected persons based on an engineered compound salience score (bounding-box area multiplied by mean keypoint confidence), and constructs a structural tensor for the skeleton processors:

```
"""
pose_extractor.py
Author: Nejra Gutić & Zeynep Nur Yılmaz
Description: Per-frame keypoint parsing and tracking of the top-2
salient subjects.
"""
import cv2
import numpy as np
from ultralytics import YOLO

class PoseExtractor:
    def __init__(self, confidence_threshold=0.25):
        # Load the native tracking model
        self.model = YOLO("yolo11n-pose.pt")
        self.conf_thresh = confidence_threshold

    def extract_coordinates(self, video_path, target_frames=150):
        cap = cv2.VideoCapture(video_path)
        frame_list = []

        while cap.isOpened():
            ret, frame = cap.read()
            if not ret:
                break
            frame_list.append(frame)
        cap.release()

        # Handle structural padding/interpolation if video length
varies
        total_frames = len(frame_list)
        sampled_indices = np.linspace(0, total_frames - 1,
target_frames, dtype=int)

        # Shape: (T, M, V, C) -> (150, 2, 17, 3)
        video_pose_data = np.zeros((target_frames, 2, 17, 3),
dtype=np.float32)

        for t_idx, f_idx in enumerate(sampled_indices):
            frame = frame_list[f_idx]
            results = self.model(frame, verbose=False)[0]

            if results.keypoints is None or len(results.keypoints) ==
0:
                continue
```

```
            detections = []
            for kpt, box in zip(results.keypoints.data.cpu().numpy(),
results.boxes.data.cpu().numpy()):
                # box structure: [x1, y1, x2, y2, conf, cls]
                bbox_area = (box[2] - box[0]) * (box[3] - box[1])
                mean_kpt_conf = np.mean(kpt[:, 2])

                # Compound salience mapping to isolate principal
interactors
                salience_score = bbox_area * mean_kpt_conf
                if mean_kpt_conf >= self.conf_thresh:
                    detections.append((salience_score, kpt))

            # Sort subjects descending based on interaction salience
            detections.sort(key=lambda x: x[0], reverse=True)

            # Retain up to the two most critical subjects
            for m_idx in range(min(2, len(detections))):
                video_pose_data[t_idx, m_idx, :, :] =
detections[m_idx][1]

        return video_pose_data
```

A.2. Skeleton and Kinematic Stream Modeling (2s-AGCN)

The skeleton stream is processed by a **Two-Stream Adaptive Graph Convolutional Network**. The following module reads the parsed keypoint matrix, computes directed bone vectors along the physical structural hierarchy of the human body, and exports graph arrays ready for MMAction2 inference pipelines:

```
"""
skeleton_stream.py
Author: Nejra Gutić & Zeynep Nur Yılmaz
Description: Dual-path configuration evaluating joint nodes and
directed bone vectors.
"""
import numpy as np
import torch
import torch.nn as nn

# Topological Kinematic Mapping for 17 COCO Keypoints
COCO_PARENTS = [0, 0, 0, 1, 2, 0, 0, 5, 6, 7, 8, 5, 6, 11, 12, 13,
14]

class SkeletonStreamProcessor:
    def __init__(self, mode="joint"):
        self.mode = mode
        self.parents = COCO_PARENTS

    def compute_bone_vectors(self, joint_tensor):
        """
        Converts absolute joint positions to relative directional
vectors.
        Input Shape: (T, M, 17, 3)
        """
```

```
        bone_tensor = np.zeros_like(joint_tensor)
        # Retain positional confidence maps across modifications
        bone_tensor[..., 2] = joint_tensor[..., 2]

        for child, parent in enumerate(self.parents):
            if child == 0:
                bone_tensor[:, :, child, :2] = 0 # Root anchor
normalization
            else:
                bone_tensor[:, :, child, :2] = joint_tensor[:, :,
child, :2] - joint_tensor[:, :, parent, :2]

        return bone_tensor

    def format_for_mmaction(self, tensor):
        """
        Permutes layout into MMAction2 convention: (M, T, V, C)
        """
        # (T, M, V, C) -> (M, T, V, C)
        formatted_tensor = np.transpose(tensor, (1, 0, 2, 3))
        return torch.tensor(formatted_tensor,
dtype=torch.float32).unsqueeze(0) # Add Batch Dim
```

A.3. Appearance and Contextual Stream Modeling (R3D-18)

The spatial-appearance representation feeds into a spatiotemporal ResNet-based network. This module implements the global random frame sampling strategy (which acts as a strong temporal regularizer) and forwards chunks through an **R3D-18** model:

```
"""
rgb_stream.py
Author: Nejra Gutić & Zeynep Nur Yılmaz
Description: Spatiotemporal ResNet feature extraction utilizing v2
random frame selection.
"""
import cv2
import random
import torch
import torch.nn as nn
import torchvision.models.video as video_models

class RGBStreamProcessor(nn.Module):
    def __init__(self, model_path=None):
        super(RGBStreamProcessor, self).__init__()
        # Initialize an R3D-18 backbone pre-trained on Kinetics-400
        self.backbone = video_models.r3d_18(pretrained=True)
        # Modify the head for binary prediction (Fight vs NonFight)
        self.backbone.fc = nn.Linear(self.backbone.fc.in_features, 2)

        if model_path:
            self.load_state_dict(torch.load(model_path,
map_dict='cpu'))
        self.backbone.eval()

    def uniform_random_sample(self, video_path, num_frames=16):
        """
```

```
        Executes whole-video random frame selection (v2 strategy
optimization)
        """
        cap = cv2.VideoCapture(video_path)
        frames = []
        while cap.isOpened():
            ret, frame = cap.read()
            if not ret:
                break
            # Standardize pixel layout and size
            frame = cv2.resize(frame, (224, 224))
            frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
            frames.append(frame)
        cap.release()

        total_frames = len(frames)
        if total_frames < num_frames:
            # Zero-padding fallback if clip is shorter than sample
size
            while len(frames) < num_frames:
                frames.append(np.zeros((224, 224, 3),
dtype=np.uint8))
            selected_frames = frames
        else:
            # Non-consecutive random global sampling
            indices = sorted(random.sample(range(total_frames),
num_frames))
            selected_frames = [frames[i] for i in indices]

        # Stack into shape: (C, T, H, W) -> (3, 16, 224, 224)
        video_tensor = np.stack(selected_frames, axis=0) # (T, H, W,
C)
        video_tensor = np.transpose(video_tensor, (3, 0, 1, 2)) # (C,
T, H, W)
        video_tensor = video_tensor / 255.0 # Normalize tracking
limits
        return torch.tensor(video_tensor,
dtype=torch.float32).unsqueeze(0)
```

A.4. Complete Pipeline Integration and Gradio Prototype Deployment

The final script brings everything together. It handles parallel forwarding through the three individual streams, computes the late-fusion scores using your grid-searched optimal weight coefficients ($w_{RGB} = 0.45$, $w_{joint} = 0.25$, $w_{bone} = 0.30$), and serves the system through a **Gradio** web panel:

```
"""
app.py
Author: Nejra Gutić & Zeynep Nur Yılmaz
Description: Unified Late-Fusion Inference Framework with a Gradio
Web Interface.
"""
import gradio as gr
import torch
import torch.nn.functional as F
```

```
import numpy as np

from pose_extractor import PoseExtractor
from skeleton_stream import SkeletonStreamProcessor
from rgb_stream import RGBStreamProcessor

# Instantiate extraction structures
extractor = PoseExtractor()
joint_processor = SkeletonStreamProcessor(mode="joint")
bone_processor = SkeletonStreamProcessor(mode="bone")

# Load fine-tuned architectures (Placeholder implementations)
rgb_model =
RGBStreamProcessor(model_path="checkpoints/r3d18_v2_best.pth")
# In production, these link directly to MMAction2 evaluation wrappers
joint_model = lambda x: torch.tensor([[0.6, 0.4]])
bone_model = lambda x: torch.tensor([[0.3, 0.7]])


def execute_late_fusion(video_path):
    """
    Combines parallel computational pathways via searched
optimization scalars.
    """
    with torch.no_grad():
       # Stream 1: RGB Spatial Appearance
        rgb_input = rgb_model.uniform_random_sample(video_path,
num_frames=16)
        rgb_logits = rgb_model(rgb_input)
        rgb_softmax = F.softmax(rgb_logits, dim=1).cpu().numpy()[0]

        # Stream 2 & 3: Coordinate Extraction and Structural Layouts
        raw_pose = extractor.extract_coordinates(video_path,
target_frames=150)

        joint_tensor = joint_processor.format_for_mmaction(raw_pose)
        bone_tensor =
bone_processor.format_for_mmaction(bone_processor.compute_bone_vector
s(raw_pose))

        # Stream Softmax Computations
        joint_softmax = F.softmax(joint_model(joint_tensor),
dim=1).cpu().numpy()[0]
        bone_softmax = F.softmax(bone_model(bone_tensor),
dim=1).cpu().numpy()[0]

        # Grid Search Calibrated Weight Distribution Coefficients
        w_rgb, w_joint, w_bone = 0.45, 0.25, 0.30

        # Matrix Weighted Accumulation (Late Fusion Formulation)
        fused_scores = (w_rgb * rgb_softmax) + (w_joint *
joint_softmax) + (w_bone * bone_softmax)

        classes = ["Non-Violence (Normal Activity)", "Violence
Detected (Alert)"]
        predicted_index = np.argmax(fused_scores)
        confidence = fused_scores[predicted_index]
```

```
        return {classes[0]: float(fused_scores[0]), classes[1]:
float(fused_scores[1])}

# Create Gradio Deployment Interface Blueprint
app = gr.Interface(
    fn=execute_late_fusion,
    inputs=gr.Video(label="Input Surveillance Footage (.mp4 /
.avi)"),
    outputs=gr.Label(num_top_classes=2, label="Late-Fusion Prediction
Probability"),
    title="Automated Violence Detection System",
    description="Three-Stream Modular Fusion Pipeline incorporating
R3D-18 Spatial Appearance and Dual 2s-AGCN Pose Skeletons.",
    theme="huggingface"
)

if __name__ == "__main__":
    app.launch(share=False)
```